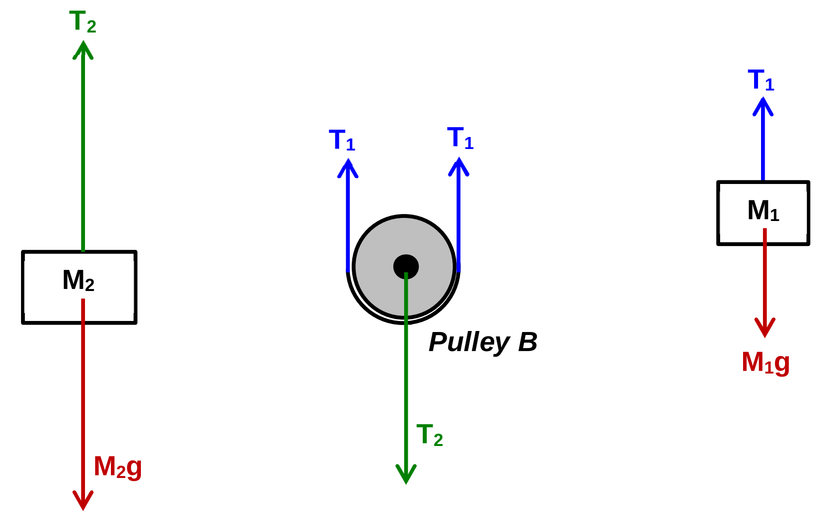
<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th></tr>
<tr><td>2(a)</td><td>The total linear momentum of a system is CONSTANT if no net external force acts on the system.</td><td>1</td></tr>
<tr><td>2(b)(i)</td><td>

$T_1 = M_1g = (4.0)(9.81) =$ 39.2 N
$T_2 = M_2g = (8.0)(9.81) =$ 78.5 N</td><td>
1
1</td></tr>
<tr><td>(b)(ii)</td><td>Since the disturbance is small and the net force on the system can be taken to be zero, hence total momentum remains at zero throughout.
$M_1v_1 + M_2(-v_2) = 0$
$v_2 = \boldsymbol{\tfrac{1}{2} v_1}$
Note that v is defined as speed in this question (i.e. no negative values for v)</td><td>1

1</td></tr>
<tr><td>(b)(iii)</td><td>The disturbance results in the increase of total kinetic energy of the system.
Energy introduced $= ½(4)(v_1^2) + ½(8)(v_2^2) =$ $\boldsymbol{3v_1^2}$</td><td>1
1</td></tr>
<tr><td>(b)(iv)</td><td>There is no change in kinetic energy when the masses were moving at constant speed.</td><td>1</td></tr>
</table>

	Lost in GPE per unit time of $M_1 = 4gv_1$ Gain in GPE per unit time of $M_2 = 8gv_2 = 4gv_1$ Hence the total energy remains the same.	1
	Max Marks	**9**

Question	Answer	Marks
3(a)(i)	P	1